MANAS REDDY ARUMALLA

Robotics & Control Engineer | MSc Autonomy Technologies, FAU Erlangen-Nürnberg

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SUMMARY

I build robots that have to work in the real world: a vision-only drone that lands itself on a moving ship deck, a bimanual arm that catches balls out of the air, a quadruped that maps and inspects a facility on its own. That work has become an IEEE publication (ICRM 2025), a hackathon win in Berlin, and 16 PRs merged upstream into Enactic's official OpenArm stack. I take ideas the whole way: derive the dynamics, design the controller, prove it in simulation, deploy it to hardware, and back every claim with tests.

EDUCATION

Friedrich-Alexander-Universität Erlangen-Nürnberg *2026 – Present*

M.Sc. Autonomy Technologies, Nürnberg, Germany

Amrita Vishwa Vidyapeetham *2021 – 2025*

B.Tech, Automation & Robotics, Coimbatore, India | CGPA 8.84/10.0 (First Class with Distinction)

Relevant coursework: Control Systems, Robotics & Mobile Robotics, Computer Vision, Machine Learning, Industrial Automation, Embedded Systems.

TECHNICAL SKILLS

Programming: Python, C++, MATLAB

Control & Estimation: LQR, MPC, SMC, PID, backstepping, H∞, adaptive & predictive control; state-space modelling; trajectory tracking (Pure Pursuit, Stanley, DWA)

Planning & Navigation: A*, RRT/RRT*, PRM, Hybrid A*, APF; SLAM (RTAB-Map); RVO multi-robot avoidance; PDDL / PlanSys2; ROS 2 Nav2

ML / RL: Reinforcement Learning (PPO), genetic-algorithm optimization, computer vision (OpenCV, MediaPipe); LLM / VLA integration

Robotics Frameworks: ROS / ROS 2, Gazebo, MuJoCo, CoppeliaSim, PX4 / MAVLink

Simulation & CAD: Fusion 360 / Inventor, Adams, Simscape, Simulink

Embedded & Hardware: Arduino, Teensy, Raspberry Pi, microcontroller programming, Embedded C, sensor integration

Operating Systems: Linux, Windows

PROFESSIONAL EXPERIENCE

Robotics Engineer, Humanoid Robotics Startup (Stealth) *Feb 2026 – Mar 2026*

- Built robotic-manipulation simulations in Gazebo and implemented PDDL task planning, letting task sequences be validated in simulation before committing to hardware.
- Simulated and tested a K-Scale humanoid in ROS 2 and stood up the sim-to-hardware deployment pipeline.

Research Assistant, Amrita Vishwa Vidyapeetham (under Dr. Rammohan Sriramdas) *Sep 2025 – Jan 2026*

- Developed a DST-funded two-wheeled hopping robot: characterized its jump dynamics and designed the stabilizing control, work published at ICRM 2025.
- Enhanced the control architecture and navigation performance of a quadruped mobile robot.
- Authored three research papers: one published, two under review.

Robotics Intern, Anvi Robotics *Jun 2024 – Jul 2024*

- Designed a water-cleaning unmanned surface vehicle (USV): component selection, CAD, and ROS simulations integrating MAVLink with PX4.
- Integrated vision-based trash detection and autonomous navigation in ROS, enabling the USV to locate and route to floating debris on its own.

Robotics Intern, Sony SSUP Program *Oct 2023 – Dec 2024*

- Co-developed a four-legged mobile robot for farm assistance and cattle health monitoring under the Sony SSUP program.
- Programmed and integrated multiple sensors and control algorithms, and troubleshot hardware to keep experiments running with minimal downtime.

Robotics Intern, 7s Technologies *Mar 2024 – Apr 2024*

- Designed an AR/VR motion-simulation chair with real-time control; selected components and ran load/performance calculations to hit motion-fidelity targets.

SELECTED PROJECTS

OpenArm: Bimanual Manipulation Platform *Python, MuJoCo*

- **Upstream contributor to Enactic's official OpenArm stack**: 16 PRs merged, 4 more in review.
- Built a control, planning, and learning stack for the **bimanual Enactic OpenArm v2 (7-DOF × 2)** in MuJoCo: forward/inverse kinematics, Cartesian and **compliant (admittance) control**, and RRT-Connect obstacle avoidance.
- Implemented a range of manipulation skills: pick-and-place, articulated-object handling (drawers/valves), cloth folding, collision-aware bimanual hand-over, and dynamic ball-catching via **Kalman + MPC interception**, plus a learned ACT vision policy and a contact-only RL suite.

Vision-Based Drone Landing on a Moving Platform *Python, MuJoCo, CasADi*

- Built a **vision-only autonomous quadrotor** that lands on moving ground rovers and 6-DOF seakeeping ship decks from onboard sensors alone: ArUco fiducials fused in a relative-state EKF with a stabilized gimbal, no ground-truth state in the loop.
- Designed the control and safety stack: **geometric SE(3)** and nonlinear / tube-DOB MPC (CasADi), a **Hamilton–Jacobi reachability safety shield**, and higher-order control barrier functions for sense-and-avoid.
- Added reinforcement-learning (PPO / LSTM-PPO) landing policies and extended the system to a **decentralized multi-drone swarm** with Kalman-consensus estimation and a CBF-QP safety filter.

panda-tamp: Closed-Loop Task & Motion Planning *Python, PDDL, YOLOv8*

- Built **closed-loop task & motion planning** for a Franka Panda in MuJoCo: PDDL + Fast Downward plans are verified against IK, grasp, and swept-path feasibility before execution, and the system replans the moment the world diverges from the plan.
- Grounded in perception: YOLOv8 detections lifted to object poses (~4 cm), **LLM goals via MCP**.

NORMA: LLM-Supervised Robot Manipulation *SmolVLA, Claude, MCP, NormaCore*

- Ran a learned VLA policy (SmolVLA) and a reasoning LLM (Claude) **together on a real 8-DOF arm**: the LLM reasons over the camera scene and supervises the policy at runtime, taking over with corrected joint commands when it stalls or fails.
- Built LLM-driven pick-and-place (no hard-coded coordinates or IK), color sorting, and gesture-based Q&A, with safety enforced by a **custom MCP server** gating every command. **Won the NormaCore track** at the HackLab Berlin AI × Robotics Hackathon.

Autonomous Quadruped Facility Inspection *ROS 2 Jazzy, Nav2, RTAB-Map, Gazebo Harmonic*

- Built a fully autonomous inspection mission for a **Unitree Go2 quadruped**: frontier exploration drives Nav2 to SLAM-map an unknown facility (RTAB-Map), the occupancy grid is segmented into rooms, and the robot visits each one with no operator input.
- Added open-vocabulary detection to 3D-localize equipment and auto-generate an inspection report; the same ROS 2 graph runs in sim and on the **real Go2 (sim-to-real)**. Built at Europe Embodied 2026 (Munich).

sparsam: SLAM & Bundle-Adjustment Backend from Scratch *C++*

- Built a sparse nonlinear-least-squares / SLAM backend from first principles: Lie groups, analytic Jacobians, block-sparse Cholesky, fill-reducing ordering, and the **Schur complement**, with no existing solver libraries.
- Benchmarked against **GTSAM and Ceres** on public SLAM / bundle-adjustment datasets (BAL Ladybug, Carlone/Grisetti suites), matching or beating solution quality, faster per iteration on larger problems.

Swarm Autonomy: Decentralized Multi-Drone Search & Pursuit *ROS 2 Jazzy, PX4 SITL, Gazebo*

- Built a **decentralized multi-UAV** system for GPS-denied search-and-pursuit where each drone runs the same onboard stack: **visual-inertial odometry (OpenVINS)**, ESDF mapping (nvblox), planning, and control, no central coordinator.
- Coordinated the swarm peer-to-peer over a modeled bandwidth-limited radio, allocating containment roles with **CBBA** to corner a fleeing target; runs full-stack on ROS 2 Jazzy + PX4 SITL + Gazebo Harmonic.

Additional: AegisDrive (uncertainty-aware autonomous-driving safety platform); BallPark (classical-vs-learned mobile-autonomy benchmark); Comparative Control Study (15+ benchmarked controllers, GA-tuned); Steady Stride (physical self-balancing robot with onboard manipulator); TurtleBot3 ROS 2 task planning (PlanSys2, PDDL, Nav2); manipdyn 6-DOF UR5e control & planning lab; quadrotor control benchmarks in Python & MATLAB.

PUBLICATIONS & RESEARCH

- M. R. Arumalla, R. Sriramdas, V. R. Devarakonda, A. Vincent, *"Mapping the Dynamics of Robotic Jumps in Hopping Robots,"* International Conference on Robotics and Mechatronics (ICRM 2025), 2025.
- Two additional manuscripts currently under review (2026).

ACHIEVEMENTS & LEADERSHIP

- **HackLab Berlin AI × Robotics Hackathon** (2026): won the NormaCore track (team of 2) for an LLM-supervised robot-manipulation system on an 8-DOF arm.
- **Europe Embodied 2026 Hackathon** (Munich): built an autonomous inspection system for a Unitree Go2 quadruped, sim-to-real deployed.
- **e-Yantra Robotics Competition 2022-23** (IIT Bombay): awarded Level 3 for an autonomous pharma-delivery bot.
- **Robotics Club**: led the projects team; designed and delivered a hands-on ROS workshop.
- **NSS Volunteer**: contributed to community camps and social-outreach initiatives.